WHEN THE INCLUSION OF THE LOCUS WHERE A SECTION IS INVERTIBLE IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14 (CONTROL 8.83 follow-up); lane-s15/carve.txt records the same edges.

Editorial note. The prefilter flagged this candidate for 'the result is clear' in the cases x in U and dim 0; those are the trivial branches of the induction and the rest of the proof is argued in full, so the card is taken under the 2026-10-14 judge-rule.

Editorial note. The proof contains one xymatrix square; it is preserved verbatim with the chapter's own Xy-pic setup.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The structure sheaf of U as a colimit of the reflexive powers of F. Lemma lemma-structure-sheaf-Xs (7456-7476), used at proof lines 7569-7579 and 7620-7624. Turns elements of Gamma(U, O_U) into sections of some F^{[n]}.
- Affineness is local on the base. Morphisms, Lemma morphisms-lemma-characterize-affine, used at proof lines 7590-7592. Reduces the converse to X affine.
- A quasi-affine scheme with vanishing higher cohomology is affine. Cohomology of Schemes, Lemma coherent-lemma-affine-if-quasi-affine, used at proof lines 7593-7595. Is the criterion applied at the end of the reduction.
- Cohomology of the inclusion commutes with flat base change. Cohomology of Schemes, Lemmas coherent-lemma-quasi-coherence-higher-direct-images-application, coherent-lemma-quasi-coherence-higher-direct-images and coherent-lemma-flat-base-change-cohomology, used at proof lines 7595-7614. Reduces the vanishing to a local ring.
- Characterisation of affine schemes by vanishing on a cover. Properties, Lemma properties-lemma-characterize-affine, used at proof lines 7628-7631. Concludes affineness of U from the affine pieces D(f_i) cap U.

Source reference. Original proposition 00KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 760–760. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EBE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17919–17919. Navigation pointer into the Stacks

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Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3372–3372. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01XJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2520–2520. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01XK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2521–2521. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0EBN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17928–17928. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01S8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2329–2329. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01QF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2264–2264. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Assumptions and notation as in Lemma 0EBN. The following are equivalent*

(1) *the inclusion morphism $j : U \to X$ is affine, and*
(2) *for every $x \in X \setminus U$ there is an $n > 0$ such that $s^n \in \mathfrak{m}_x \mathcal{F}_x^{[n]}$.*

Proof. Assume (1). Then for $x \in X \setminus U$ the inverse image U_x of U under the canonical morphism $f_x : \mathrm{Spec}(\mathcal{O}_{X,x}) \to X$ is affine and does not contain x. Thus $\mathfrak{m}_x \Gamma(U_x, \mathcal{O}_{U_x})$ is the unit ideal. In particular, we see that we can write

$$1 = \sum f_i g_i$$

with $f_i \in \mathfrak{m}_x$ and $g_i \in \Gamma(U_x, \mathcal{O}_{U_x})$. By Lemma 0EBN we have $\Gamma(U_x, \mathcal{O}_{U_x}) = \mathrm{colim}\, \mathcal{F}_x^{[n]}$ with transition maps given by multiplication by s. Hence for some $n > 0$ we have

$$s^n = \sum f_i t_i$$

for some $t_i = s^n g_i \in \mathcal{F}_x^{[n]}$. Thus (2) holds.

Conversely, assume that (2) holds. To prove j is affine is local on X, see Morphisms, Lemma 01S8. Thus we may and do assume that X is affine. Our goal is to show that U is affine. By Cohomology of Schemes, Lemma 0EBE it suffices to show that $H^p(U, \mathcal{O}_U) = 0$ for $p > 0$. Since $H^p(U, \mathcal{O}_U) = H^0(X, R^p j_* \mathcal{O}_U)$ (Cohomology of Schemes, Lemma 01XK) and since $R^p j_* \mathcal{O}_U$ is quasi-coherent (Cohomology of

Schemes, Lemma 01XJ) it is enough to show the stalk $(R^p j_*\mathcal{O}_U)_x$ at a point $x \in X$ is zero. Consider the base change diagram

$$\begin{array}{ccc} U_x & \longrightarrow & U \\ {\scriptstyle j_x}\downarrow & & \downarrow{\scriptstyle j} \\ \mathrm{Spec}(\mathcal{O}_{X,x}) & \longrightarrow & X \end{array}$$

By Cohomology of Schemes, Lemma 02KH we have $(R^p j_*\mathcal{O}_U)_x = R^p j_{x,*}\mathcal{O}_{U_x}$. Hence we may assume X is local with closed point x and we have to show U is affine (because this is equivalent to the desired vanishing by the reference given above). In particular $d = \dim(X)$ is finite (Algebra, Proposition 00KQ). If $x \in U$, then $U = X$ and the result is clear. If $d = 0$ and $x \notin U$, then $U = \emptyset$ and the result is clear. Now assume $d > 0$ and $x \notin U$. Since $j_*\mathcal{O}_U = \mathrm{colim}\, \mathcal{F}^{[n]}$ our assumption means that we can write

$$1 = \sum f_i g_i$$

for some $n > 0$, $f_i \in \mathfrak{m}_x$, and $g_i \in \mathcal{O}(U)$. By induction on d we know that $D(f_i) \cap U$ is affine for all i: going through the whole argument just given with X replaced by $D(f_i)$ we end up with Noetherian local rings whose dimension is strictly smaller than d. Hence U is affine by Properties, Lemma 01QF as desired. □